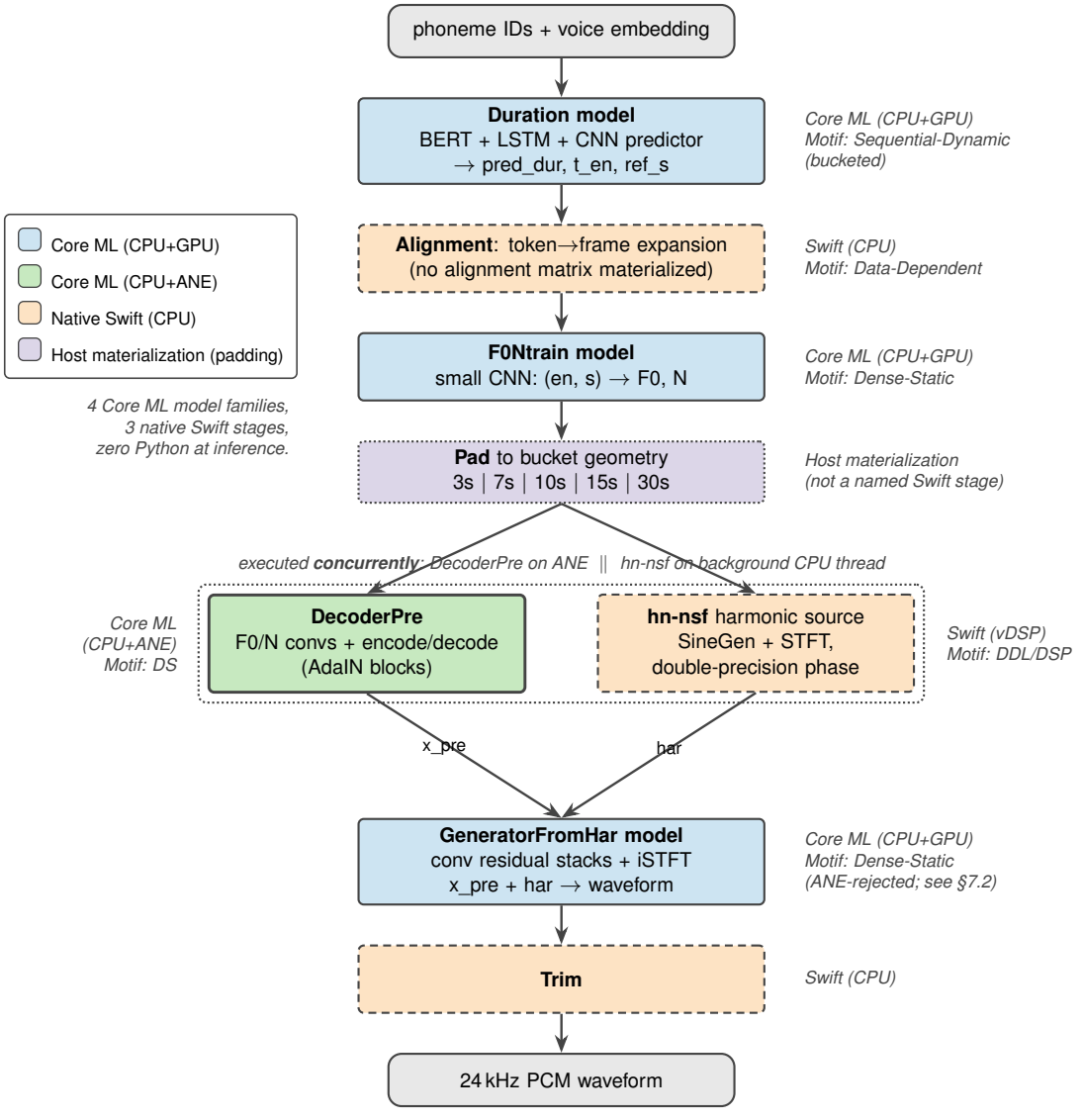
phoneme IDs + voice embedding
Duration model
BERT + LSTM + CNN predictor
→ pred_dur, t_en, ref_s
Core ML (CPU+GPU)
Motif: Sequential-Dynamic
(bucketed)
Core ML (CPU+GPU)
Core ML (CPU+ANE)
Native Swift (CPU)
Host materialization (padding)
4 Core ML model families,
3 native Swift stages,
zero Python at inference.
Alignment: token→frame expansion
(no alignment matrix materialized)
Swift (CPU)
Motif: Data-Dependent
F0Ntrain model
small CNN: (en, s) → F0, N
Core ML (CPU+GPU)
Motif: Dense-Static
Pad to bucket geometry
3s | 7s | 10s | 15s | 30s
Host materialization
(not a named Swift stage)
executed concurrently: DecoderPre on ANE || hn-nsf on background CPU thread
Core ML
(CPU+ANE)
Motif: DS
DecoderPre
F0/N convs + encode/decode
(AdaIN blocks)
hn-nsf harmonic source
SineGen + STFT,
double-precision phase
Swift (vDSP)
Motif: DDL/DSP
x_pre
har
GeneratorFromHar model
conv residual stacks + iSTFT
x_pre + har → waveform
Core ML (CPU+GPU)
Motif: Dense-Static
(ANE-rejected; see §7.2)
Trim
Swift (CPU)
24 kHz PCM waveform